

AHAN CHAKRABORTY

Kolkata, West Bengal, India
Curriculum Vitae
chakraborty.ahan0@gmail.com

EDUCATION

- **B.Stat** 2024 – present
Indian Statistical Institute, Kolkata
• Percentage: 97.73(till present)
• University Rank : 1 (till present)
- **Higher Secondary Education** 2022 – 2024
Ramakrishna Mission Boys' Home High School, Rahara
• Percentage: 97.6
• Secured a rank of 9 in the state (out of $\sim 7,60,000$ students)
- **Secondary Education** 2012 – 2022
Ramakrishna Mission Boys' Home High School, Rahara
• Percentage: 97.2
• Secured a rank of 13 in the state (out of $\sim 11,00,000$ students)

EXPERIENCE / PROJECTS

- **A Reading Project on Random Graphs, Expectation Thresholds**
ISI Kolkata
• **Supervisor** : Prof. Soumendu Sundar Mukherjee, ISI Kolkata
• **Brief Description** : Explored probabilistic concepts, including branching processes, coupling, the probabilistic method, phase transitions, connectivity, sub-critical and super-critical regimes, and subgraph containment thresholds in Erdős-Rényi random graph from standard texts (Alon-Spencer, Remco Hofstad, Karonski-Frieze, Roch, Rick Durrett). Concluded with expectation thresholds and the Park-Pham Covering Theorem and Kahn-Kalai Conjecture.
- **A Reading Project on Convex Clustering**
ISI Kolkata
• **Supervisor** : Prof. Saptarshi Chakraborty, University of Michigan
• **Brief Description** : Conducted a comprehensive reading project on traditional clustering techniques and Sparse Hierarchical, Convex, Biconvex, Sparse Convex clustering methods.
- **An Analysis of Habitability of Exo-planets using Learning Tools**
ISI Kolkata
• **Supervisor** : Prof. Smarajit Bose, Prof. Asis Chatterjee, Prof. Amita Pal, ISI Kolkata
• **Brief Description** : Extracted key habitability features and predicted the habitability of exoplanets utilizing machine learning tools.
- **Structural and Exchange Mobility as Additive Components of the Transition Matrix**
ISI Kolkata
• **Supervisor** : Prof. Debasis Sengupta, ISI Kolkata
- **A Reading Project in Statistical Learning**
ISI Kolkata
• **Supervisor** : Prof. Smarajit Bose, ISI Kolkata

- **Brief Description :** Explored statistical learning tools, linear programming, and convex optimization through foundational texts (e.g., ESL, ISLR, Statistical Pattern Recognition by Fukunaga, Convex Optimisation by Boyd and Linear Programming by Matousek). Additionally, reviewed research papers on Generalized QDA and hub-awareness in KNN classifiers.
- **2025 Summer School in Statistics for Astronomers**
Center for Astrostatistics at The Pennsylvania State University

ACADEMIC ACCOLADES

- Secured Rank 5 All India, in Madhava Mathematics Competition, 2025
- University topper in Simon Marais Mathematics Competition, 2024, rank 13 in East Division, 4 in India, in pair, with Arnab Sanyal, received Jane Street cash prize
- Received prize money for good academic performance in B.Stat First Year
- Secured rank 12 in B.stat Entrance 2024, ISI Kolkata(out of ~15,000 candidates)
- Received SHRIRAM Scholarship in CMI Entrance, 2024 (rejected, ~50 students were given the scholarship all over India)
- Secured All India Rank 1227 in JEE Advanced (IIT), 2024 (out of ~1.8 lakhs candidates)
- Secured Rank 15 in WBJEE, 2024 (out of ~1 lakh candidates)
- Qualified Regional Math Olympiad, 2023, attended INMOTC 2024
- Received *Jagadis Bose National Talent Search (JBNSTS) Junior Scholarship*, 2022 funded by Department of Science and Technology and Biotechnology, Government of West Bengal

ACADEMIC INTERESTS

- Statistical Learning
- Probability Theory
- Mathematical Statistics
- Combinatorial Game Theory

TALKS

- At Math Club ISIK on a probabilistic proof of Cayley's Theorem using Poisson Branching Process

SKILLS

- Programming: R, C, $\text{\LaTeX}$, HTML
- Website: ahanchak2005.github.io

ADDITIONAL INFORMATION

- **Hobbies:** Singing, Active trainee in Indian Raga Music, Learning languages, Watching movies, Reading novels, Learning instruments
- **Extra Curricular Activities:** Member of Core Committee in Math Club, ISI Kolkata, and Cultural Committee, ISI Kolkata
- **Languages:** Bengali (native), Hindi, English (professional proficiency), and Italian(elementary proficiency)